

BULBUL SAINI

Full-Stack Software Engineer • Java & Web Development

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EDUCATION

Women Engineering College — *Bachelor of Technology in Computer Science & Engineering*
Cumulative GPA: 8.02 / 10.0 (Ranked Top 10% in Department)

Expected May 2026
Ajmer, Rajasthan

Aryan International Academy — *Senior Secondary (Class XII, RBSE)*
Academic Merit: 92.2% in Physics, Chemistry, and Mathematics

May 2021
Rajasthan, India

EXPERIENCE

DEE Piping Systems — *Software & Full-Stack Developer Trainee*
Full-Stack Engineering & Application Development

Jun 2024 – Present
India

- **Selected** to spearhead the front-end architecture of **12+ reusable UI components** using **React.js** and **Node.js**, decreasing client render time by **28%**.
- **Coordinated** with backend engineers to deploy **8+ RESTful API endpoints** with JSON validation, processing **1,000+ daily requests** with 99.9% uptime.
- **Refactored** core service layers applying architectural design patterns (MVC, Factory), eliminating **25% redundant code** across database access layers.
- **Collaborated** with a cross-functional team of 5 engineers in bi-weekly Agile/Scrum sprints, conducting **15+ code reviews** to achieve 100% on-time release.

Birla Institute of Scientific Research — *Java Development Intern*
Applied Research & Software Quality Assurance

May 2024 – Jul 2024
Jaipur, Rajasthan

- **Orchestrated** Java algorithm optimizations using the **Java Collections Framework** (HashMaps, PriorityQueues), reducing search complexity from O(N) to O(1) and accelerating query speed by **35%**.
- **Partnered** with senior researchers to construct **5+ modular class hierarchies** adhering to SOLID principles, reducing memory overhead by **18%**.
- **Automated** test verification procedures and executed **50+ manual QA test cases** alongside QA leads, identifying and resolving **15+ defects** prior to launch.

TECHNICAL PROJECTS

Full-Stack Authentication & Authorization Platform — *TypeScript, React.js, Node.js, SQL*

Aug 2024 – Oct 2024

- **Architected** an end-to-end authentication system supporting **500+ active user accounts** with registration, session management, and RBAC security.
- **Deployed JWT (JSON Web Tokens)** with HTTP-only cookies and **bcrypt** password hashing (10 salt rounds), securing **100% of sensitive API routes** against XSS and CSRF vulnerabilities.
- **Formulated** normalized **MySQL** schemas with indexed primary and foreign keys, decreasing credential lookup query latency by **40%**.
- **Constructed** responsive client interfaces in **TypeScript** and **React.js** with real-time field validation, reducing invalid form submissions by **30%**.

Movie Ticket Booking Simulation System — *Java, Multithreading, MVC Architecture, Data Structures*

Jan 2024 – Mar 2024

- **Engineered** a multi-screen cinema booking application managing **10+ movie catalogs**, **50+ showtimes**, and concurrent seating grids.
- **Implemented MVC architectural design** across 3 decoupled layers, achieving **100% modular separation** between UI and business logic.
- **Standardized** real-time seat matrix locking using **2D arrays** and **Java HashMaps**, eliminating **100% of double-booking conflicts** under simulated peak loads.
- **Integrated** structured exception-handling covering **12+ boundary edge cases**, guaranteeing zero runtime crashes during transaction simulations.

TECHNICAL SKILLS

Programming Languages: Java, JavaScript, TypeScript, Python, C, SQL

Web & Backend Frameworks: React.js, Node.js, Express.js, HTML5, CSS3, RESTful APIs, JSON, Responsive Web Design

Databases & Developer Tools: MySQL, Relational Database Design, Database Indexing, Git, GitHub, Docker, Postman, VS Code

Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming (OOP), Java Collections Framework, DBMS, Operating Systems

LEADERSHIP & CERTIFICATIONS

- **Academic & Peer Leadership:** Mentored 15+ junior engineering students in Data Structures, Java OOP, and software design patterns.
- **Java Software Architecture & Algorithms Specialization** — Coursera
- **Full-Stack Web Development & REST API Design** — LinkedIn Learning